

RQ Answers

omitted

2024-04-05

Data overview

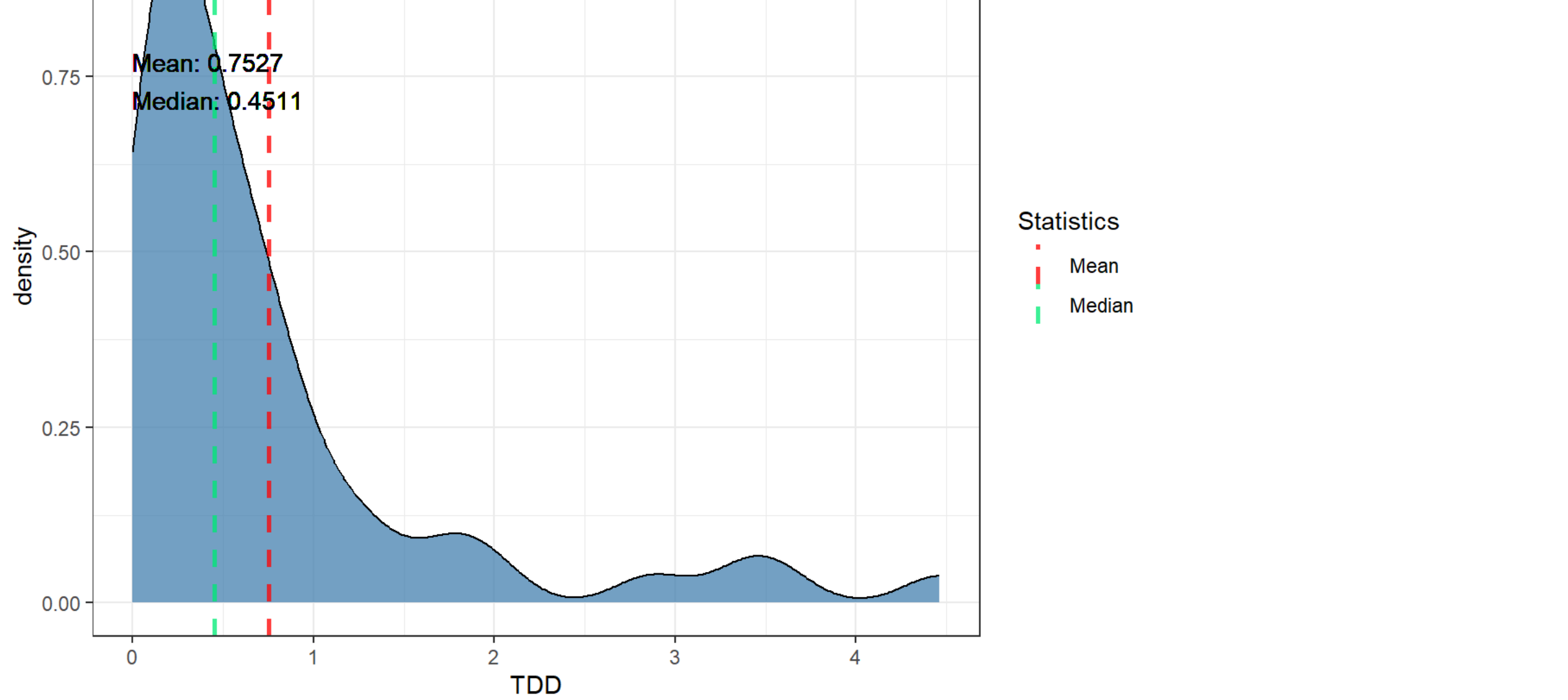
glimpse(all_data)				
## Rows: 1,605				
## Columns: 11				
## \$ repo	<chr> "opennlp", "opennlp", "opennlp", "opennlp", "opennlp", "opennlp", "opennlp", "opennlp", "opennlp", "opennlp", "opennlp"			
## \$ pr_number	<int> 153, 197, 207, 216, 216, 216, 216, 216, 240, 240, ...			
## \$ rule	<chr> "java:S4973", "java:S112", "java:S112", "java:S1192", "java:S18...			
## \$ severity	<chr> "MAJOR", "MAJOR", "CRITICAL", "MAJOR", "MINOR", "C...			
## \$ file	<chr> "opennlp-tools/src/test/java/opennlp/tools/util/St...			
## \$ type	<chr> "BUG", "CODE_SMELL", "CODE_SMELL", "CODE_SMELL", "..."			
## \$ status	<chr> "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "OPEN", "O...			
## \$ debt	<int> 5, 20, 10, 15, 5, 12, 2, 2, 8, 8, 5, 12, 10, 20, 2...			
## \$ ncloc_affected_file	<int> 60, 23, 117, 32, 32, 32, 32, 64, 64, 3, 132, 1...			
## \$ complexity	<int> 7, 4, 12, 5, 5, 5, 5, 5, 5, 0, 18, 18, 18, ...			
## \$ origin	<chr> "NEW", "NEW", "PRE-EXISTING", "NEW", "NEW", "PRE-E...			

RQ1

TDD (Technical Debt Density) by PR Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdd), : All aesthetics have length 1, but the data has 52 rows.
## | Please consider using `annotate()` or provide this layer with data containing a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdd), : All aesthetics have length 1, but the data has 52 rows.
## | Please consider using `annotate()` or provide this layer with data containing a single row.
```

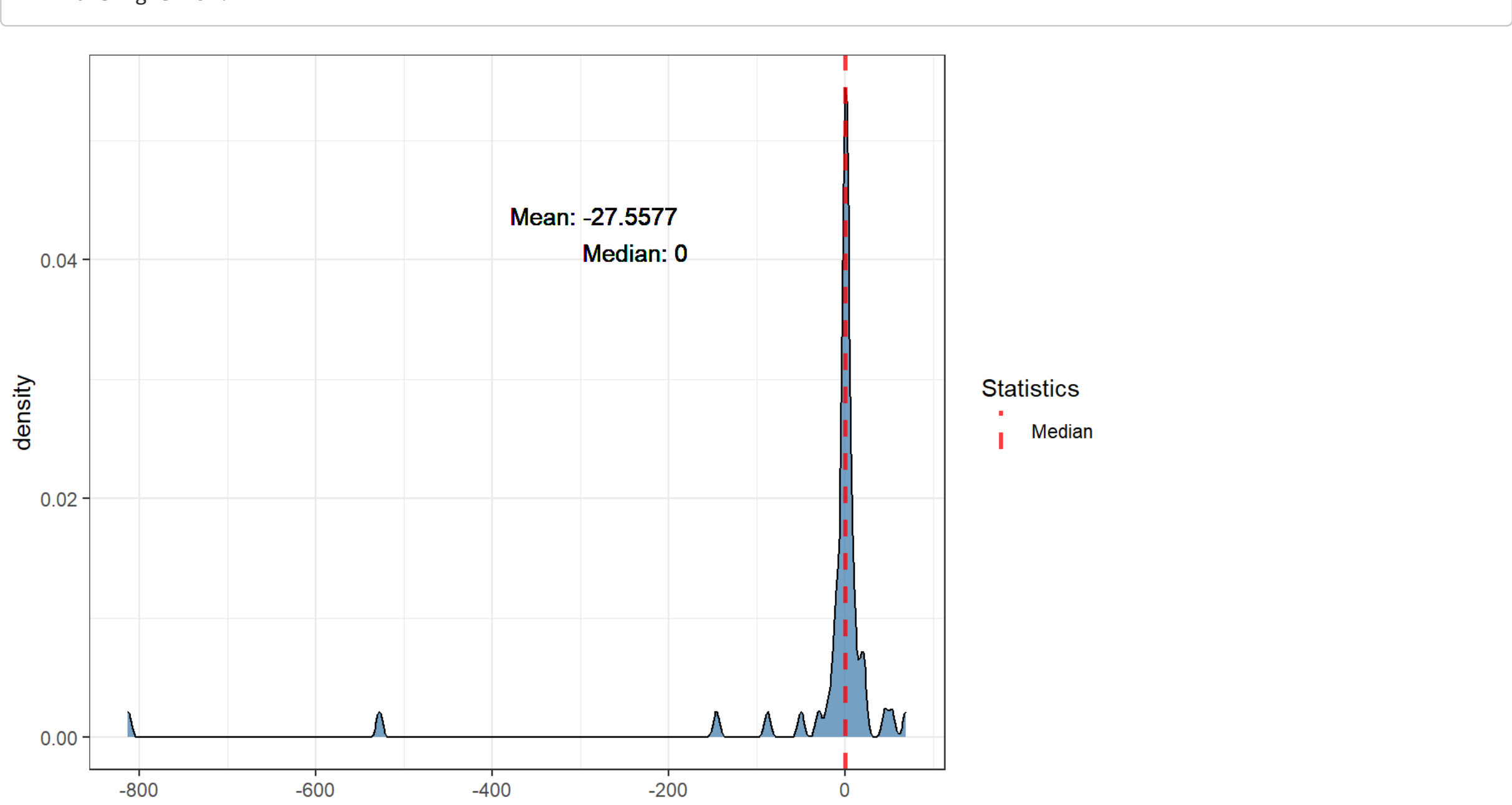


TDV (Technical Debt Variation)

TDV Distribution

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Mean:", round(mean(tdv), : All aesthetics have length 1, but the data has 52 rows.
## | Please consider using `annotate()` or provide this layer with data containing a single row.
```

```
## Warning in geom_text(aes(x = 0, y = 0, label = paste("Median:", round(median(tdv), : All aesthetics have length 1, but the data has 52 rows.
## | Please consider using `annotate()` or provide this layer with data containing a single row.
```



Percentages for TDV < 0 (TD decrease), TDV = 0 (TD unchanged) and TDV > 0 (TD increased) by repo

## # A tibble: 1 × 4				
## repo	tdv_lower_than_zero	tdv_equal_to_zero	tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	<dbl>	
## 1 opennlp	26.9	38.5	34.6	

Mean of percentages of TDV (mean of repo percentages)

## # A tibble: 1 × 3			
## mean_tdv_lower_than_zero	mean_tdv_equal_to_zero	mean_tdv_greater_than_zero	
## <dbl>	<dbl>	<dbl>	
## 1	26.9	38.5	34.6

Percentages of pre-existing TDV by repo

## # A tibble: 1 × 3			
## repo	preexisting_tdv_equal_to_zero	preexisting_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	
## 1 opennlp	64.6	35.4	

Mean of percentages of pre-existing TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## preexisting_tdv_equal_to_zero	mean_preexisting_tdv_greater_than_zero	
## <dbl>	<dbl>	
## 1	64.6	35.4

Percentages for new TDV by repo

## # A tibble: 1 × 3			
## # Groups: repo [1]			
## repo	new_tdv_equal_to_zero	new_tdv_greater_than_zero	
## <chr>	<dbl>	<dbl>	
## 1 opennlp	96.2	3.85	

Mean of percentages of new TDV (mean of repo percentages)

## # A tibble: 1 × 2		
## mean_new_tdv_equal_to_zero	mean_new_tdv_greater_than_zero	
## <dbl>	<dbl>	
## 1	96.2	3.85

RQ2

As data is unbalanced across repositories, as shown in the issue characterization (`./issues_characterization.Rmd`), we will examine the top 10 fixed issues and the top 10 unfixed issues for each repository, and then aggregate them into a rank using the *PositionScore* metric given by the following formula:

$$PositionScore_i = \sum_{j=1}^k Position_{ij}$$

given a rule i , its *PositionScore_i* will be the sum of its position across all k repositories. After calculating the metrics for all rules, we rank them in a TOP 10 of the most frequent rules (fixed or unfixed, depending on the context) across repositories. The lower the *PositionScore*, the more frequent the rule across repositories.

TOP 10 fixed per repo ranked by count

All top 10 fixed rules for each repo.

## # A tibble: 10 × 2		
## # Groups: rank [10]		
## rank	opennlp	
## <int>	<chr>	
## 1	1 java:S1854	- 54
## 2	2 java:S112	- 33
## 3	3 java:S1192	- 32
## 4	4 java:S1130	- 26
## 5	5 java:S1764	- 23
## 6	6 java:S2095	- 18
## 7	7 java:S3008	- 13
## 8	8 java:S108	- 8
## 9	9 java:S2200	- 8
## 10	10 java:S1116	- 8

TOP 10 fixed

TOP 10 fixed rules by PositionScore metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S1854	122	
## 2 java:S112	123	
## 3 java:S1192	124	
## 4 java:S1130	125	
## 5 java:S1764	126	
## 6 java:S2095	127	
## 7 java:S3008	128	
## 8 java:S108	129	
## 9 java:S1116	130	
## 10 java:S2200	131	

TOP 10 unfixed per repo ranked by count

All top 10 unfixed rules for each repo.

## # A tibble: 10 × 2		
## # Groups: rank [10]		
## rank	opennlp	
## <int>	<chr>	
## 1	1 java:S1854	- 373
## 2	2 java:S1192	- 269
## 3	3 java:S106	- 96
## 4	4 java:S2095	- 85
## 5	5 java:S1602	- 66
## 6	6 java:S1135	- 35
## 7	7 java:S112	- 27
## 8	8 java:S1168	- 26
## 9	9 java:S1874	- 26
## 10	10 java:S1186	- 20

TOP 10 unfixed

TOP 10 unfixed rules by PositionScore metric.

## # A tibble: 10 × 2		
## rule	position_score	
## <chr>	<dbl>	
## 1 java:S1854	122	
## 2 java:S1192	123	
## 3 java:S106	124	
## 4 java:S2095	125	
## 5 java:S1602	126	
## 6 java:S1135	127	
## 7 java:S112	128	
## 8 java:S1168	129	
## 9 java:S1874	130	
## 10 java:S1186	131	

Outliers

Higher TDD

## # A tibble: 52 × 5				
## repo	pr_number	total_debt	total_ncloc	tdd
## <chr>	<int>	<int>	<int>	<dbl>
## 1 opennlp	563	1035	232	4.46
## 2 opennlp	569	1161	325	3.57
## 3 opennlp	585	1296	386	3.36
## 4 opennlp	559	665	232	2.87
## 5 opennlp	455	70	35	2
## 6 opennlp	487	69	38	1.82
## 7 opennlp	518	167	99	1.69
## 8 opennlp	423	160	118	1.36
## 9 opennlp	443	75	59	1.27
## 10 opennlp	216	36	32	1.12
## # i 42 more rows				

Higher TDV

## # A tibble: 52 × 3			
## pr_number	repo	tdv	
## <int>	<chr>	<int>	
## 1	487 opennlp	69	
## 2	569 opennlp	53	
## 3	585 opennlp	45	
## 4	252 opennlp	22	
## 5	197 opennlp	20	
## 6	216 opennlp	20	
## 7	522 opennlp	15	
## 8	420 opennlp	10	
## 9	421 opennlp	10	
## 10	465 opennlp	10	
## # i 42 more rows			

Lower TDV

## # A tibble: 52 × 3			
## pr_number	repo	tdv	
## <int>	<chr>	<int>	
## 1	431 opennlp	-813	
## 2	559 opennlp	-528	
## 3	466 opennlp	-146	
## 4	524 opennlp	-88	
## 5	502 opennlp	-50	
## 6	486 opennlp	-30	
## 7	471 opennlp	-20	
## 8	543 opennlp	-15	
## 9	414 opennlp	-10	
## 10	533 opennlp	-10	
## # i 42 more rows			

Extra

Number of PRs with pre-existing TD

## n	
## 1 301	

Number of java:S1192 issues that affect test code

## n	
## 1 0	

Number of java:S1192 issues that affect not test code

## n	
## 1 0	

Summary NCLOC

## ncloc	
## Min. : 18.00	
## 1st Qu.: 72.25	
## Median : 172.50	
## Mean : 480.85	
## 3rd Qu.: 281.00	
## Max. : 9275.00	